

Nihal Machhi

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EDUCATION

Parul University

Bachelor of Technology in Computer Science and Engineering(AI/ML) | CGPA – 7.07

Vadodara, GJ

Sep. 2025 – May 2028

Parul University

Diploma in Information Technology | CGPA – 7.53

Vadodara, GJ

Sep. 2022 – May 2025

PROJECTS

[DPI-Engine](#) | Python, FastAPI, Scapy, Jinja2, WebSockets, Uvicorn, Computer Networks

- Built real-time DPI engine processing 10K+ packets/sec via multi-threaded consistent-hash pipeline
- Implemented protocol fingerprinting via TLS SNI, HTTP Host, and DNS across 50+ live flows
- Designed producer-consumer pipeline with sentinel shutdown, reducing latency via zero-copy buffers
- Validated engine with 34-test suite covering TCP reassembly and malformed packets

[MeshPay](#) | Python, FastAPI, Uvicorn, Cryptography, Pydantic, Pytest, HTTPX

- Designed offline UPI protocol using public-key crypto for secure internet-free P2P payments
- Built QR-based handshake and multi-device relay with signed receipts blocking double-spending
- Wrote Pytest/HTTPX suite covering verification, request handling, and failure edge cases
- Built async FastAPI endpoints to minimize payment-confirmation round-trip latency

[TrackitNow](#) | React, Vite, FastAPI, PostgreSQL, Supabase, Chart.js, Cloudinary, JWT

- Built full-stack habit tracker with 15+ tasks across 4 categories on React/FastAPI/PostgreSQL
- Engineered 12-month contribution heatmap using Chart.js and custom aggregation queries
- Secured 10+ REST endpoints with JWT auth and bcrypt, sanitizing input against XSS/SQL injection
- Built real-time chat and friend system on Supabase changefeeds for concurrent messaging

EXPERIENCE

Placement coordinator Intern | Parul University

Aug 2024 – May 2025

- Coordinated 20+ recruitment drives by managing student-company data and placement workflows
- Collaborated with 15+ recruiters and placement teams, supporting 80+ student placements
- Maintained and updated placement databases, ensuring accurate student and recruiter records

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript, TypeScript, SQL

Full Stack Engineering: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs

AI & Machine Learning: PyTorch, Hugging Face, LangChain, Pinecone, OpenCV, Gemini API

Cloud & DevOps: MongoDB, PostgreSQL, Firebase, Firestore, GCP, Supabase, Docker

System Tools & Platforms: Git/GitHub, Linux/Unix, Postman, Vercel, Streamlit

ACHIEVEMENTS

- **Secured Runner-Up at KananHack 2026** (Kanan x Parul University) among 24 competing teams in a 24–48hr sprint
- **Built prototypes across 5 hackathons** — Google The Big Code, Odoo x Parul University, and HackerRank Orchestrate
- Solved 500+ DSA problems on LeetCode (arrays, graphs, trees, DP), reaching contest rating 1400+
- Applied Computer Networks and cryptography coursework to build DPI-Engine and MeshPay
- Coordinated placement operations for 100+ students as Placement Coordinator, managing 20+ recruitment drives
- Completed Anthropic's Claude API and Claude Code certifications, applying LLM tool-use and agentic workflows